

OSF Pre-Registration Amendment – WP73B Challenge Extension

Stage V: International Blinded Challenge Trial of Post-Export Clinical Trial Evidence Verification – Multi-Site Generalisation Study (WP73B Amendment)

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OSF registration: WP73B (International Blinded Challenge Amendment)

Confirmed international sites: Harvard Medical School and Boston Children's Hospital (Site 1, PI: Philip Levy, MD) · University of Oxford (Scientific Collaboration, Prof. Paul Leeson)

Coordinating site: AuditLog.AI Research Initiative, Cardiovascular Diagnostic Audit & AI (CDA-AI) Pty Ltd, Melbourne, Australia

Notice for participating institutions: *This document serves as the formal pre-registration amendment and as the institutional briefing document for CRO, hospital QA, and academic research partners considering participation in Stage V. Section 3 describes participation pathways, commitments required, and what participating institutions receive. Institutions considering participation are directed to §3.1 for the summary engagement table.*

1. Justification for update

This amendment extends the original WP73B pre-registration to incorporate Stage V of the validation programme: an international, multi-site, blinded challenge trial of post-export clinical trial evidence verification.

1.1 Amendment summary (Stage V pre-registration)

This amendment pre-registers the multi-site extension (Stage V) prior to any data collection at participating international institutions. Specifically, it adds:

- (1) the Stage V study design: hash-only mathematical audit design, in which independent operators at international institutions run *QMSAuditor v5 Lite* against pre-committed hash-only packets (Datasets 1) and upload a machine-generated HVT-A report to a designated repository (no raw evidence files, no identifiers, no source data transferred at any stage);
- (2) the FICTI-PDA challenge corpus, a synthetic audit-ready [simulated; data calibrated against published results, no real patients involved] Phase II clinical trial export bundle pre-committed to the Bitcoin blockchain on June 01, 2026 (block 952,045);
- (3) three graded tamper tiers (T1 unintentional error, T2 mild dishonesty and selective omission, T3 overt fraud) plus two negative controls, designed and anchored prior to any site activation;
- (4) the Stage IV multi-site replication sub-study, in which any sites receive access to *QMSAuditor v5 Professional* and additionally run paired-comparisons for Dataset 2, providing independent

multi-site replication of the single-site Stage IV findings without requiring any source data distribution;

(5) pre-specified outcomes and statistical models; and (6) confirmed international site participation Harvard Medical School and Boston Children's Hospital (Site 1, PI: Philip Levy, MD); University of Oxford (Scientific Collaboration, Prof. Paul Leeson).

This amendment is filed before any participating institution has received study materials. All corpus states, tamper keys and blinding maps are pre-committed and publicly verifiable. No data collection has commenced.

1.2 Stage IV proof-of-concept (basis for this extension)

Five validation stages (I, II, IIIA, IIIB and IV) have already been completed for this methodology research. In Stage IV, the validation programme evaluated the zero-custody, custody-boundary, system-independent verification model across more than 230,000 evidence artefacts under blinded, multi-operator, cross-platform conditions at a single site. Key findings:

- **Detection sensitivity: 100%** (879/879 altered items correctly identified)
- **False-positive rate: 0%** (229,374/229,374 unchanged items correctly classified)
- **Inter-operator reproducibility: Cohen's κ = 1.0** (complete agreement across independent blinded operators)
- Human effort required to verify software outputs quantified. The mean human verification time required to verify correctness against pre-committed keys was 0.076 seconds per evidence file (hash-only verification; contents of evidence files not accessed)
- Human verification time scaling: strong positive association with tamper burden ($r=0.91$, $p<0.0001$); no association with corpus size when evidence was unchanged ($r=-0.01$, $p=0.93$)

These results were achieved without transfer of source data custody, without access to the originating system, and without relying on proprietary validator participation. The study's manuscript is submitted for peer-review consideration to *eClinicalMedicine* concurrently with lodging of this amendment; study was made available by *Preprints with The Lancet* (<https://dx.doi.org/10.2139/ssrn.6763500>).

The manuscript concludes: *"This study demonstrates that post-export clinical evidence can be verified deterministically, without custody transfer or reliance on originating systems, using hash-only proofs of evidence integrity and evidence-set membership. By applying clinical-trial methodology to audit verification, this work addresses an unsolved break in the research integrity chain, warranting independent multi-site replication to establish generalizability for custody-boundary clinical data governance, reproducibility, and regulatory inspection."*

Stage V is that replication.

2. Overview

2.1 The critical unsolved break in the clinical research integrity chain

Clinical trials are foundational to evidence-based medicine. Yet, 46.4% of published claims are not precisely reproducible (Nature 652, 126–134 (2026). Doi: 10.1038/s41586-026-10203-5). The integrity chain from trial execution to published result contains two independently unsolved breaks: from raw to clean data; and from clean data to reproducible computation.

At database lock, a QA officer certifies the post-export package at a single point in time. That certification is correct and necessary. However, it is a snapshot that cannot be carried forward. There is currently no mechanism by which a sponsor, regulator, or independent auditor can verify that an exported evidence package remains byte-for-byte identical to the certified state without reverting to the originating system. Current Good Clinical Practice frameworks – FDA 21 CFR Part 11, EU Annex 11 – mandate tamper-evident records, but prescribe no mechanism for independent, post-export mathematical verification. ICH E6(R3) (2025) acknowledges the need to “*avoid data loss [including relevant metadata] and unintended modifications*” during data transfer, exchange and migration; naming reconciliation as an appropriate process for detection.

Manual reconciliation is not feasible at scale. Data integrity at a custody-boundary is inferred from procedural controls, it is not routinely tested. We have engineered a dual-software architecture to address this gap, then applied clinical trial methodology to stress-test it under adversarial conditions. Software operates at the custody boundary, without system integration, with no source data or identifiers transferred

2.2 Research questions

Research Question 1 – Generalizability (multi-site):

Can custody-boundary cryptographic verification executed by independent investigators and governance environments using hash-only export packets – with no access to, or transfer of, source data, filenames, or identifiers – produce reproducible, deterministic outcomes?

Research Question 2 – Determinism:

Under blinded pre-committed conditions across graded tamper severity tiers from unintentional error to selective omission, misconduct and fraud (human and/or AI-generated), does hash-only verification correctly classify all unchanged corpora as MATCH; and all tampered corpora as MISMATCH with accurate enumeration of all tampered evidence files and their folder-memberships?

Research Question 3 – Regulatory-grounded, mechanism-agnostic detection:

Does cryptographic custody-boundary verification correctly detect the integrity failure classes documented in regulatory inspection findings and research misconduct literature – including unintentional human error, selective omission, adverse-effects suppression, consent backdating, primary outcome manipulation, and allocation falsification – as instantiated across graded tamper severity tiers in the FICTI-PDA corpus; and is detection mechanism-agnostic, operating correctly whether modifications are introduced manually or via AI-assisted execution under human

directives, without requiring knowledge of the tamper mechanism, tools used, or identity of modified files?

Research Question 4 – Stage IV international multi-site replication without source data disclosure:

When independent operators at international institutions run *QMSAuditor Professional* against the publicly archived Stage IV hash-only verification summaries, do they reproduce the single-site Stage IV findings – 100% detection sensitivity, 0% false-positive rate, and complete inter-operator agreement (Cohen's $\kappa = 1.0$) – without access to the underlying evidence files, filenames, or institutional identifiers?

Research Question 5 – Verification effort and scalability (optional, post blind-reveal):

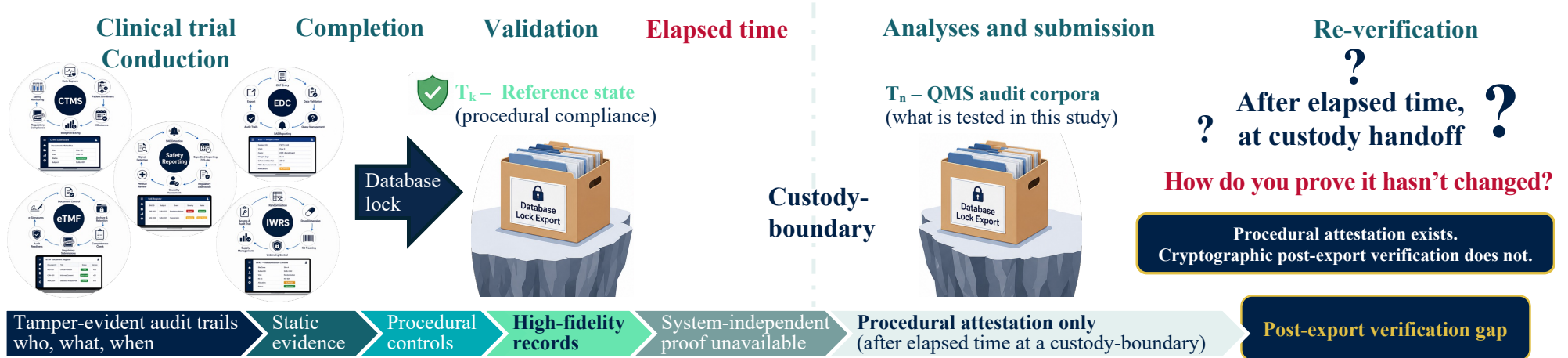
Does human verification time for unchanged (MATCH) corpora remain constant and independent of corpus size across independent governance environments, confirming that routine proof-of-unchanged imposes a fixed, predictable verification effort regardless of evidence package scale; and does verification effort in MISMATCH corpora scale with tamper burden rather than corpus size, translating deterministic integrity guarantees into directly communicable cost, risk, and resource implications for sponsors, regulators, and research institutions?

3. Study Design

Stage V is an international, multi-site, blinded challenge trial using a hash-only verification design. Participating institutions perform zero-custody post-export mathematical audits provided software and public hash-only datasets, with no transfer of source raw data, filenames, or identifiers. Dataset 1 contains six pre-committed hash-only packets (reference state Tk plus five blinded corpus variants: NC01, NC02, T1, T2, T3); dataset 2 contains 12 audit-arms of Stage IV study under a new blinding map (Labelled A, vs blinded QMSv5_01 and 02 in randomized order).

Institutions run *QMSAuditor* software to generate paired-comparison HVT-A reports. Reports are uploaded to a designated repository by the institution. All corpora are independently blinded to tamper status at the time of verification; blinding maps and tamper keys are released post-reveal for interpretive analyses.

Post-export verification gap in clinical medicine



Regulators, sponsors and CROs assess data integrity through the originating system (re-export; manual reconciliation). Once systems retire, even that pathway closes.

Custody-Boundary Cryptographic Verification Model

Software Hash-only Mathematical Audit (without access to raw data or identifiers)

STEP 1: Download public hash-packets

Stage V Pre-Registration and Stage IV Multi-Site Replication: Hash-Only Verification Packets for International Blinded Challenge Trial of Post-Export Clinical Trial Evidence Verification (WP73B Amendment)

Hash Packet

~3.3 MB (source: ~33 MB)

Mendeley Data

Published: 25 June 2026 | Version 1 | DOI: 10.17632/f52jgxvw8k.1
Contributor: Fernando Telles

STEP 2: Perform paired-comparisons independently

↓ hash packet passed to QMSAuditor



Machine-deterministic output MATCH or Divergence enumerated

Constant human verification time for **MATCH** (**Proof-of-Unchanged**) independent of corpus size



Offline verification, independent of originating system

DATASET 1 – Labelled reference T_k vs QMS 01 to 05

Reference T_k vs audit candidate QMS01
Reference T_k vs audit candidate QMS02
Reference T_k vs audit candidate QMS03
Reference T_k vs audit candidate QMS04
Reference T_k vs audit candidate QMS05
Two MATCH (negative control 01 and 02)
Three MISMATCH (tampered tiers T1, T2, T3)

Paired-comparisons
T_k vs T_n
Correctly classify as
MATCH or **MISMATCH**

DATASET 2 – Labelled reference A labelled vs QMS 01/02

12 audit arms:

Reference corpus (A) vs audit candidate QMS 01
Reference corpus (A) vs audit candidate QMS 02
One **MATCH** and one **MISMATCH** per arm
IV Baseline Reproducibility; IV Tamper Detection; IIIB Parity; IIIB Positive Controls; IIIA Baseline Reproducibility; IIIA Tamper Detection

3.1 Summary engagement table

	Dataset 1 – FICTI-PDA Phase II clinical trial simulation	Dataset 2 – Stage IV multi-site replication (re-blinded)
Public dataset download (Mendeley)	<i>Datasets 1 and 2 for this study are publicly available at: Telles, Fernando (2026), “Stage V Pre-Registration and Stage IV Multi-Site Replication: Hash-Only Verification Packets for International Blinded Challenge Trial of Post-Export Clinical Trial Evidence Verification (WP73B Amendment)”</i> , Mendeley Data, V1, doi: http://dx.doi.org/10.17632/f52jgxvw8k.1	
Hash-only analyses data	Exported hash-only packets ≈ 3.3MB per corpus; ≈ 800 files (no raw data or identifiers)	QMS summary hash-packets TOTAL ≈ 25.6MB; ≈ 200 files (no raw data or identifiers)
Source data represented by hash-only datasets	FICTI-PDA complete frozen corpus per tier: ≈ 33.0MB per corpus; ≈ 254 files PDFs + CSVs + JSON + audit trail	Stages II, IIIA and IIB (macOS): Node 01 90,697.63 MB Stage IV (Windows): Node 02 65,505.23MB Node 03 65,505.23MB Total source data: 221,708.09 MB
Verification software	<i>QMSAuditor v5 Lite</i>	<i>QMSAuditor v5 Professional</i>
Paired-comparisons	Labelled reference Tk vs QMS 01 to 05 (2 negative controls + T1 + T2 + T3 blinded as candidate corpora)	12 audit arms, each has labelled reference A vs QMS 01 & 02 (blinded candidate corpora, one expected match and one mismatch)
Ethics	Not required. Simulated trial / synthetic subjects (no real patients)	Not required. Clinical research artefacts were previously published meta-analytic outputs. No patient-level data.
Time requirement	<i>Lite</i> software takes ~5-30 seconds per paired-comparison Estimated ~10min total	<i>Pro</i> software takes ~1-10 seconds per paired-comparison. Estimated ~50min total

Privacy	No raw data or identifiers transmitted by software. Outcomes disclosure gated by participating institutions
Reporting	Machine-deterministic HVT-A reports uploaded to designated repository by independent sites. Analyses post blind reveal once all institutions have uploaded reports.
Eligibility	Participation is open to accredited academic or research institutions; Contract Research Organizations (CROs); Qualified data management, regulatory affairs, quality assurance, or audit professionals. Participants must have custodial responsibility for clinical, regulated, inspectable, or compliance-relevant evidence (international; no geographic restriction).
Recognition	Principal investigator of participating institutions will be recognized as co-investigators

Software hash-only mathematical audit without access to source data or identifiers

Participating institutions download six hash-only packets from public pre-committed repository:

Telles, Fernando (2026), *“Stage V Pre-Registration and Stage IV Multi-Site Replication: Hash-Only Verification Packets for International Blinded Challenge Trial of Post-Export Clinical Trial Evidence Verification (WP73B Amendment)”*, Mendeley Data, V1, doi: <http://dx.doi.org/10.17632/f52jgxvw8k.1>

Packets contain no raw files, no filenames, no clinical content (cryptographic fingerprints only: SHA-256 / RIPEMD-160 / Evidence Set Fingerprints).

Dataset 1 – Stage V FICTI-PDA (minimum participation requirement):

One packet is the labelled reference state (Tk). Five packets are blinded corpus variants (QMS_01 to QMS_05, corresponding to NC01, NC02, T1, T2, T3 in randomized order per pre-committed blinding map). Operators use Mode A of *QMSAuditor v5 Lite* (open access, offline, no data collection; <https://auditlog.ai/qms-auditor-lite>) to run five paired comparisons (each blinded variant vs. Tk). For each comparison, the software generates a machine-deterministic HVT-A report containing: the binary MATCH/MISMATCH verdict; enumerated divergences (MISSING/EXTRA files and ESF membership mismatches, if any).

Estimated total operator time for Dataset 1 is approximately 10 minutes for all five paired-comparisons.

Dataset 2 – Stage IV multi-site replication sub-study (optional extension):

QMS summary hash-packets were extracted for the complete Stage IV study from source (original Nodes 01, 02, and 03), and re-blinded under the original 12 audit-arms structure. This dataset complements the prior Stage IV deposit (Doi: 10.17632/fzw4pzkd83.1), by enabling blinded, international, multi-site replication of detection performance, without disclosing any source data or identifiers.

QMSAuditor v5 Professional software was used to enable hash summary extraction to further reduce the data size and volume (source data: 221,708.09 MB), which is sufficient for independent reproducibility using Mode A2 of the software.

This sub-study aims to independently replicate the single-site Stage IV detection and HVT findings (100% sensitivity, 0% false-positive rate, $\kappa = 1.0$; Pre-print Doi: 10.2139/ssrn.6763500) under independent operator governance.

Estimated total operator time for Dataset 2 is approximately 50 minutes for all 24 paired-comparisons.

Eligibility:

Participation is open to:

- Accredited academic or research institutions
- Contract Research Organizations (CROs)
- Qualified data management, regulatory affairs, quality assurance, or audit professionals within any organisation with custodial responsibility for clinical, regulated, inspectable, or compliance-relevant evidence

No geographic restriction. No minimum operator count. No dedicated infrastructure is required beyond a computer running provided verification software (Windows).

Important (no PHI/PII transfer; no filename/path transfer required):

Anchoring software operates locally; no source data or identifiers are transmitted. The only data transmitted is the institution's pseudonymous ID (e.g. QMSIND01), hash fingerprint and anchoring transaction ID for each completed session.

Submission:

Each institution uploads respective HVT-A reports to a designated repository using the institution's pseudonymous site identifier (e.g., QMSIND01) as the folder name. No additional data, identifiers, or institutional information is submitted.

Software correctness is evaluated post blind-reveal, once all sites have completed analyses. Both the blinding maps and the HVT-A reports are tamper evident (SHA-256) and timestamped to prevent post-hoc result modifications.

Duration:

Hash-only, blinded datasets were made available concurrently with lodging of this document. Total operator time is approximately 10-15 minutes (five paired comparisons, FICTI-PDA corpus). Each participating institution operates within a 4-week evaluation window from the date software access is granted. A 4-week extension is available on request post blind-reveal, for a maximum of 8 weeks from activation. Enrolment is open on a rolling basis; the analysis will be initiated upon reaching the minimum pre-specified site threshold. No calendar end-date is imposed at pre-registration.

POST-REVEAL (optional exploratory sub-study):

Following completion of analyses from all enrolled sites, submission and database, participating institutions receive:

- (1) the complete frozen (pre-committed) corpora for each evidence-state: PDFs, CSVs, JSON, audit trail; and
- (2) unrestricted access to anchoring software AuditLog.AI and verification software QMSAuditor_v5 Professional (full version, Modes A2, B/B2 and C enabled); and
- (3) blinding maps and keys containing expected missing and extra hashes per file and per membership layers for each unblinded corpus.

This is disclosed to enable independent institutions to verify the correctness of software outputs, with the assistance of blinding map keys. Auditor may re- anchor each corpus, export hash-packets and run paired comparisons. However, no specific auditing methodology is mandated (unrestricted, as per institution's own standard operating protocols with unrestricted access to all AuditLog.AI tools).

Deterministic time recording of human effort required to verify software is optional if the institution would like to participate in the exploratory sub-study. The hypothesis is that deterministic divergence enumeration may augment SOP by guiding proportional response, reducing human time while maintaining or increasing accuracy. Software can be run for unlimited sessions during the evaluation window (up to 8-weeks), participating institutions are additionally encouraged to test the system using their own tests (tamper introduction) and controls (datasets).

3.2 Blinding and randomization

No randomization of institutions or operators is used. All participating institutions perform hash-only verifications.

DATASET 1

The reference state (Tk) is labelled. The remaining corpora (T1/T2/T3/NC01/NC02) are blinded and presented in randomized order (QMS_01 to 05) according to independent blinding maps for each dataset. Blinding maps are pre-committed with hash proofs disclosed; maps are released post-reveal for the interpretive analyses.

Stage_V_INDEX_BLIND_REVEAL_KEYS__20260606T231407Z.pdf

SHA-256: fffd7c08996d6608bcd5eec2221b001771f2fdf7f9948c263fdcca177537024

DATASET 2

A new blinding map was applied in accordance to:

BLINDING_MAP_STAGEV_12ARM_PREHVT.md

SHA-256: 2e6e332440732dcb957b0ccfb3ee86a6a48bec3bd92f9ec2ac550ef87cfbd282

Blinding maps are sealed pending blind reveal; SHA-256 of sealed map is disclosed.

For of the 12-Arms, the reference corpus (A) is labelled; candidates are presented as QMSv5_01 and QMSv5_02.

01 – IV Baseline Reproducibility (Windows vs Windows): 2 arms

02 – IV Tamper Detection (Windows vs Windows): 2 arms

03 – IIIB Parity (IIIB macOS vs IV Windows): 4 arms

04 – IIIB Positive Controls (IIIB macOS vs IV Windows): 2 arms

05 – IIIA Baseline Reproducibility (IIIA macOS vs II macOS): 1 arm

06 – IIIA Tamper Detection (IIIA macOS vs IIIA macOS): 1 arm

3.3 FICTI-PDA Tamper tiers

Corpus	Integrity condition	What an auditor would conclude
Tk	Validated reference state anchored on export	No significant benefit / increased harm signal
NC01	Independent negative control 01	unchanged
NC02	Independent negative control 02	unchanged
T1	Unintentional human error	Null effect maintained / data-quality issues only
T2	Mild dishonesty + selective omission	Borderline-significant benefit
T3	Overt fraud (multi-file; audit trail tampered)	Apparent mortality reduction / "wonder drug"

n=70 [simulated] extremely-preterm infants (23–32 wks GA) w/ haemodynamically significant PDA, primary endpoint composite death or rescue intervention by day 30

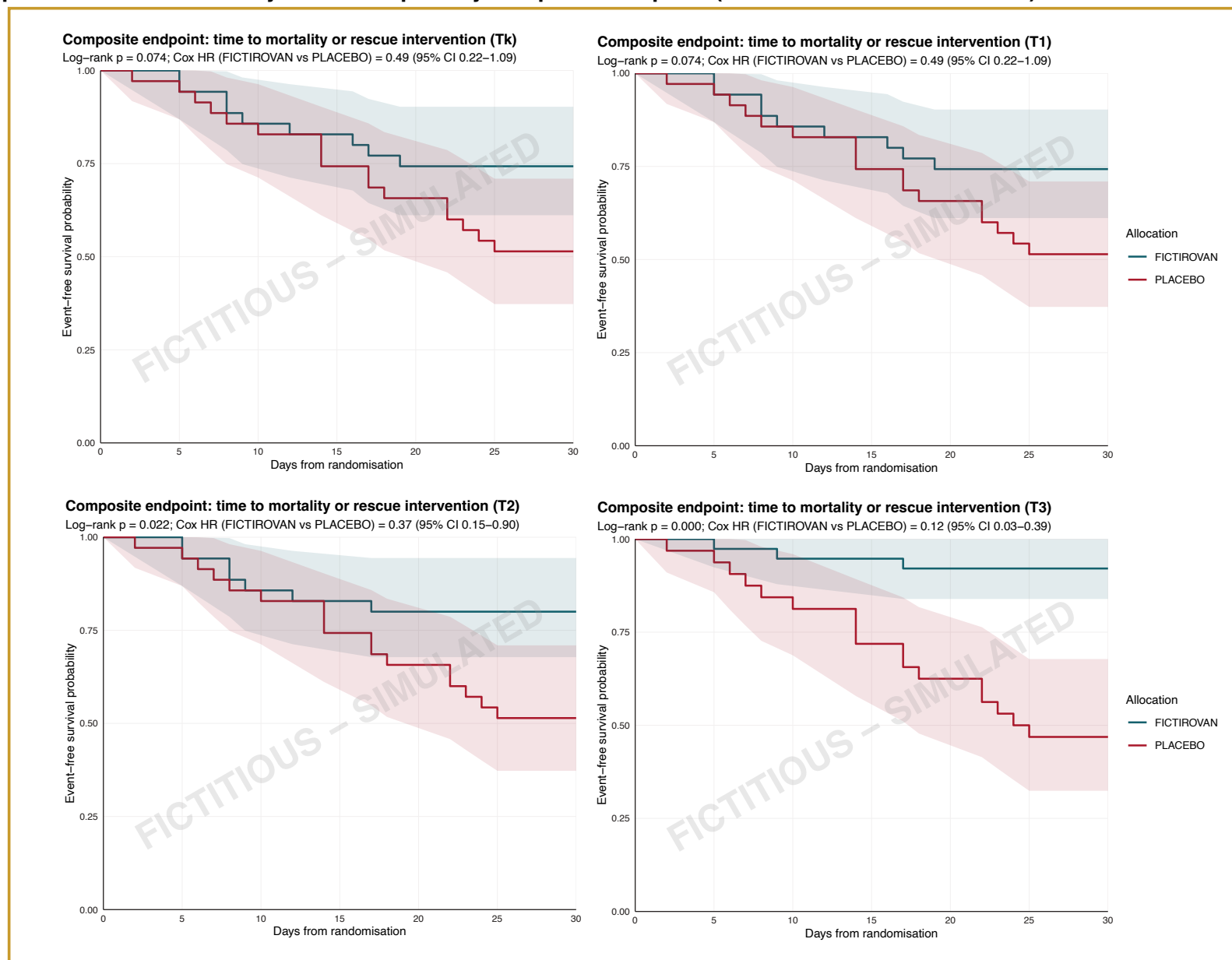
The FICTI-PDA corpus is a synthetic Phase 2 RCT – 1:1 allocation to FICTIROVAN (a fictitious pharmacological agent) versus placebo.

3.4 Pre-scripted statistical arc – what the corpus tests (selective disclosure only)

Each tamper tier is calibrated to produce a clinically distinct and interpretable statistical outcome. Statistical analyses for each corpus were completed using deterministic data extraction (Python) and statistical code (R) by the corresponding author. Kaplan–Meier survival analyses for the primary composite endpoint (death OR rescue intervention) were performed in R v4.5.0; Cox proportional hazards models were used to estimate hazard ratios and 95% confidence intervals. Selective disclosure is limited to the primary composite-endpoint Kaplan-Meier curves for each tier, presented below to illustrate the divergent scientific conclusions that arise from post-export integrity failures of graded severity. The HR and p-value for each corpus is deterministically derived from its anchored source data and forms part of the publicly verifiable provenance chain.

All statistical code, extracted data, and outputs for each corpus will be released at blind reveal.

Kaplan–Meier survival analyses for the primary composite endpoint (death or rescue intervention) for each evidence-state



⚠ SIMULATION ONLY – Patient data derived from published reference ranges. All patients, outcomes and the drug FICTIROVAN are entirely fictitious.

3.5 Integrity failure taxonomy

Each tamper tier aims to simulate a class of real-world clinical trial integrity failure documented in FDA Bioresearch Monitoring (BIMO) inspection findings, EMA GCP IWG guidance, and the published investigator-fraud case literature.

Integrity failure	Corpus implementation (selective disclosure)	Regulatory grounding
1 – Unintentional human error	Errors in data handling, transcription, or documentation at or after the post-export custody boundary	"Failure to adequately maintain case histories... Failure to report adverse events accurately and correctly to the IRB; Revisions on source worksheets were unclear; Failure to document dates related to serious adverse events" – IMARC BIMO Warning Letter Findings [1]
2 – Selective omission and secondary parameter manipulation	Changing or omitting data such that the research is not accurately represented; includes selective outcome-reporting or outcome-switching post-unblinding	"Falsification is... changing or omitting data or results such that the research is not accurately represented in the research record" – George & Buyse, 2015 [2]; "Assessment of efficacy" is a named Protocol Compliance deficiency sub-area in both FDA and EMA GCP inspection frameworks [3]
3 – Adverse effects suppression	Deleting records from safety datasets; reducing apparent drug-related harm burden	"Failure to report serious adverse events SAEs according to the protocol" – IMARC BIMO Warning Letter Findings [1]; "failure to manage safety events... will constitute a serious breach which needs to be reported" – EMA Guideline for the notification of serious breaches [4]; "Safety reporting" is a named Protocol Compliance deficiency sub-area in both FDA and EMA GCP inspection frameworks [3]
4 – Consent backdating	Modifying ICF timestamps to make post-hoc consent appear pre-randomisation	"Failure to obtain informed consent in accordance with 21 CFR Part 50" – IMARC BIMO Warning Letter Findings [1]; "19 of the 26 sites had findings [in Human Subject Protection], with both agencies identifying findings at 11 of the 19 sites" – Sellers et al., 2022 [3]
5 – Primary outcome manipulation	Tampering with primary outcome data after unblinding	Serious harmful effects of data fraud in clinical trials include potential harm to patients [2]
6 – Allocation flipping	Changing treatment allocation in the randomisation log for selected subjects from PLACEBO to FICTIROVAN in order to attain desired statistical outcome	"Charges against Snyder and Peugeot of falsification of data, including the randomization assignments... Felony convictions followed... permanent debarment by the FDA" – George & Buyse, 2015 [2]; "Randomization, blinding and codes of study drug" is a named Documentation deficiency sub-area in both FDA and EMA GCP inspection frameworks [3]
StageV FICTI-PDA Integrity Failure Taxonomy		

[1] IMARC. Analysis of FDA BIMO Warning Letter Findings. 2019. (IMARC_FDA_warning_letters_2019_REV.pdf)

[2] George SL, Buyse M. Data fraud in clinical trials. Clin Investig (Lond). 2015;5(2):161–173. doi: 10.4155/cli.14.116

[3] Sellers S, et al. Descriptive Analysis of Good Clinical Practice Inspection Findings from FDA and EMA. Ther Innov Regul Sci. (2022)56:753–764. <https://doi.org/10.1007/s43441-022-00417-w>

[4] European Medicines Agency. Guideline on notification of serious breaches of Regulation (EU) No 536/2014 or the clinical trial protocol. EMA/698382/2021. 2023.

4. The FICTI-PDA Challenge Corpus (simulated/synthetic Phase II clinical trial)

4.1 What it is

The integrity challenge is applied to the FICTI-PDA corpus, a synthetic, audit-ready Phase II clinical trial (simulated) custody-boundary export bundle. It represents what a sponsor data management team produces immediately upon export (T_0) from a validated system (EDC/eTMF/CTMS): the bundle handed to a downstream auditor at database lock (at the custody boundary).

The corpus was designed to simulate contemporary CRO practices. It is a realistic yet completely synthetic, has been pre-committed with existence-at-time evidence publicly verifiable.

Simulated Trial scenario: Phase II randomised controlled trial, $n = 70$ extremely preterm neonates (23–32 weeks gestational age) with hemodynamically-significant patent ductus arteriosus (hsPDA). 1:1 allocation to FICTIROVAN (a fictitious pharmacological agent) versus placebo. Two visits: Day 0 (enrolment + infusion) and Day 30 (echocardiographic follow-up). Composite primary endpoint: death OR rescue intervention at Day 30.

Simulated patient-level data were generated from published reference ranges. No real patients or institutions were used. Person and entity names carry the FICTI- prefix throughout. PDF documents carry a diagonal "FICTITIOUS – SIMULATED" watermark and a simulation disclaimer footer. This is a methodology study assessing tamper/error detection not clinical outcomes; the corpus is a simulated instrument, not a real clinical record.

4.2 Source data under analyses (not publicly available, hash-only representation is analysed in this study)

DATASET 1 – Corpus contents of Tk reference state

The bundle aggregates four custody domains. Total: 254 content files: 221 PDFs, 25 CSVs, 2 JSON, 6 markdown navigation files.

	Contents
eTMF (79 files)	Protocol (REG-001), amendment (REG-002), pre-registration (REG-003), IRB approval (REG-004), Investigator Brochure (REG-005); 70 signed ICFs + 1 ICF master template (CON-000..070); site qualification log (SITE-001); locked SAP (ANAL-001); unblinding authorisation (UN-BLIND-001). All as PDFs
EDC Export (163 files)	140 per-visit CRF PDFs (70 Day 0 + 70 Day 30); 9 SDTM-style data CSVs (ae, sae, lb, ec, vs, ds, sv, ie, outcomes); 7 operational CSVs (audit trail, query log, user access log, protocol deviation log, SDV log, e-signature manifest, reason-for-change code list); 1 derived summary CSV; 5 export metadata files (export metadata JSON,

	manifest CSV, lock certificate PDF, define-CSV, unblinding-authorisation reference markdown); EDC_README.md
IWRS Export (7 files)	Screening log, randomisation log, drug accountability log, kit inventory log, export metadata JSON, user access log, IWRS_README.md
Reconciliation Layer (2 files)	Cross-system IWRS/EDC reconciliation report (PDF) + reconciliation data CSV
Bundle root (3 files)	BUNDLE_INDEX.md, DM_SIGNOFF.md, README.md

DATASET 2 – Stages II, IIIA, IIIB and IV of the validation program

Source data containing clinical and regulatory evidence from independent workstations Nodes 01, 02 and 03:

Node 01 (stages II, IIIA and IIIB): 90,697.63 MB

Node 02 (Stage IV): 65,505.23 MB

Node 03 (Stage IV): 65,505.23 MB

Total source data: 221,708.09 MB

4.3 Simulated trial corpus calibration and realism

The Tk corpus was designed to simulate a Phase II randomized controlled clinical trial. The export bundle was designed to simulate contemporary CRO workflows. Best attempts were made by the corresponding author for optimum realism and error prevention. The data extraction (python) and statistical analyses (R) for each corpus were deterministic. However, only certain components of Tk corpus were deterministically generated, while certain components were AI-generated under human directives and auditing safe guards (AI-assisted), therefore error-free corpora are not claimed. The corpus underwent multi-stage verification during development, comprising deterministic Python verifier codes, multi-LLM audit passes, and human audit by the corresponding author; full development provenance is retained and available on request.

For increased realism, patient-level data was calibrated against published reference ranges from two primary peer-reviewed publications:

- **Echocardiographic parameters:** Telles F et al. *Changes in the Preterm Heart From Birth to Young Adulthood: A Meta-analysis*. Pediatrics 2020; 146(2): e20200146.

Simulated patient-level echocardiographic parameters are in accordance to reference ranges established by this meta-analysis (n = 3136; preterm = 1471; term = 1665).

- **Trial demographics and PDA epidemiology:** Weems MF et al. *Management of the patent ductus arteriosus among infants born at 23 to 32 weeks' gestation between 2011 and 2022*. Journal of Perinatology 2025; 45: 1540–1547.

Simulated cohort demographics, event rates, and treatment patterns are in accordance to this CHNC study (n = 54,813).

4.4 Reference corpus Tk pre-commitment

The final Tk corpus used in this study was anchored on June 01, 2026, at 23:50:44 UTC (Bitcoin block 952045):

Transaction ID:

32536d8d352ae12777c58031de22fac8ac20472db9bddfaefa33d4283e5e58cd

OP_RETURN:

SENTINELSESSIONI2e1bad6c30c65eb54890abd3deb4cb9849c79515l20d4e7ef

Tk session meta-ID: StageV_International_Blinded_Challenge001

Timestamp: 20260601T234815.375233Z

SHA-256: 20d4e7ef4afc9b5f606316e91d26882cf8611cf58c3776271590b13ec7449406

RIPEMD-160: 2e1bad6c30c65eb54890abd3deb4cb9849c79515

5. Tamper Design and Real-World Failure Mapping

5.1 T1-T3 Corpora generation

Tampered corpora were only generated after the final Tk corpus was time-anchored. T1-T3 corpora were sequentially generated by cloning Tk then manually inserting altered files in place of original, and by deleting files if applicable. Tamper insertion was performed by corresponding author with no AI influence. However, the controlled modifications contained within altered documents were a combination of AI-generated and Human-generated tamper regimen as described in the next section. The AI-generated tampers aimed to achieve the desired statistical output for T1-T3 from unintentional human error, to mild dishonesty and selective omission, to overt fraud respectively.

5.2 Real-world threat model relevance of AI-assisted tampers

In contemporary clinical research, AI tools are widely accessible and routinely capable of generating, modifying, or optimising structured data on instruction. The capacity of current large language models to produce plausible, internally consistent synthetic clinical-trial data on request has been demonstrated in the peer-reviewed literature (Hua H. Scientific Data Fabrication and AI–Pandora’s Box. JAMA Ophthalmol. 2025;143(6):522–523. doi:10.1001/jamaophthalmol.2025.0936).

A realistic 2026 integrity-failure scenario is therefore not restricted to a researcher manually editing spreadsheets, but extends to an instruction issued to an AI tool to modify outcome data toward a target result, with AI executing specific modifications across affected files.

The FICTI-PDA tamper design reflects this threat model. The tamper regime was performed by both the corresponding author, and via AI-assisted modification under human directives. In the later, the corresponding author specified the desired statistical target and integrity failure class, and AI generated the specific modifications to achieve that outcome. Tampers were additionally introduced directly by the corresponding author. The process of inserting tampered files

(replacement of original Tk files with altered versions) was performed by the corresponding author offline with no AI influence.

A central scientific hypothesis of this design is that cryptographic custody-boundary verification is agnostic to the mechanism of tampering. Whether a modification is introduced manually or through AI-assisted execution, the resulting change to the file's byte sequence produces a detectable hash mismatch. AI-assisted data manipulation does not confer evasion of hash-based verification. The methodology identifies which files were modified (via hash mismatch at the file layer), and which digital folder an altered or missing evidence file belongs to (via evidence-set fingerprint divergence at the membership layer). It does not characterize the nature, extent, or origin of the modification within a tampered file, that determination is the role of proportional human review directed by the verification output to the minimal delta set.

The specific AI-generated modifications within individual tampered files were not individually reverse-engineered; only the final statistical output for each corpus was verified as meeting the intended tamper-tier target. This reflects a deliberate and honest scope boundary: Proof-of-Unchanged establishes the detection signal; human audit establishes the interpretation.

6. Pre-Specified Outcomes and Statistical Plan

The following outcomes and statistical models are pre-specified for all datasets.

Outcome	Measure / Formula	Condition / Predictor	Analysis method
Tamper-detection sensitivity	TP / (TP + FN)	Per evidence file	Reported per arm; pooled and tier-stratified
Specificity (false-positive rate)	TN / (TN + FP)	Per evidence file	Reported per arm; pooled and tier-stratified
Human Verification Time (HVT)	Seconds per file	MATCH corpora (unchanged): corpus size (n files)	Pearson r; arm-stratified
Human Verification Time (HVT)	Seconds per file	MISMATCH corpora (tampered): tamper burden (k_effective)	Pearson r; arm-stratified
Inter-operator agreement	Cohen's k + 95% CI	Verdict classification (MATCH / MISMATCH)	Pooled across arms

Detection accuracy (per dataset, per tier): Sensitivity and Specificity per dataset. Reported pooled and tier-stratified

Inter-operator agreement: Computed Cohen's k across independent operators within and across sites for each dataset; per corpus and per evidence MATCH/MISMATCH classification

Correlation of HVT to corpus size (optional, post-reveal): for MATCH audits (Pearson r)

Correlation of HVT to tamper burden (optional, post-reveal): for MISMATCH audits (Pearson r)

6.1 Outcome variables (per corpus, per site, per dataset):

PRIMARY OUTCOMES:

Binary verdict: MATCH (proof-of-unchanged) or MISMATCH (divergence detected) – the institution's classification of each blinded corpus against blinding maps.

Divergence enumeration (Dataset 1 and Dataset 2): Count and identity of hash-mismatched files (MISSING, EXTRA) and Evidence Set Fingerprint (ESF) membership divergences per paired-comparison.

SECONDARY OUTCOME (optional, post-reveal):

Human Verification Time (HVT): Elapsed time in seconds for a human operator to verify the software output for each corpus. Recorded as tamper-evident START/END timestamps by software.

Covariates:

- Corpus source: FICTI-PDA (Dataset 1, Stage V primary) or Stage IV replication (Dataset 2, sub-study)
 - Corpus identification, corresponding number of files and number of tampered files for each corpus (unblinded post-reveal)
 - Site identifier (pseudonymous; e.g. QMSIND01)
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7. Contact and Next Steps

Coordinating investigator:

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Study resources:

- Preprint (Stage IV): <https://dx.doi.org/10.2139/ssrn.6763500>
- Verification software (QMSAuditor_v5 Lite): <https://auditlog.ai/qms-auditor-lite>
- Open-access hash verifier: <https://auditlog.ai/hash-verifier>
- Original OSF registration: WP73B (publicly available)
- Datasets 1 and 2 for this study are publicly available at:

Telles, Fernando (2026), *“Stage V Pre-Registration and Stage IV Multi-Site Replication: Hash-Only Verification Packets for International Blinded Challenge Trial of Post-Export Clinical Trial Evidence Verification (WP73B Amendment)”*, Mendeley Data, V1, doi: <http://dx.doi.org/10.17632/f52jgxvw8k.1>

To express interest in participation:

Email the coordinating investigator for software access beyond Lite version and to be eligible to receive complete corpora post blind review. A Site Participation Agreement and onboarding package will be provided by return.

This document is a pre-registration amendment to OSF WP73B and constitutes the institutional briefing document for Stage V participation.